

AYUSH RANJAN ROY

AI Engineer specializing in production multi-agent systems, LLM governance, and security-hardened cloud infrastructure.

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EXPERIENCE

Full Stack LLM Development Analyst

Aug 2025 - Present

Accenture, GenAI & Data, Gurugram | Multi-Agent Systems & LLM Infrastructure

- Eliminated duplicate state processing and approval failures, measured by **zero unhandled errors in production**, by co-designing a **stateful cyclic FSM** in LangGraph with **OAuth 2.0** ingestion, DB-backed queue (queueinv), **CTE stored procedures**, Postgres-checkpointed **HITL** interrupts, and optimistic-locking duplicate prevention.
- Took O2C document processing from PoC to **300+ docs/day at 97%+ accuracy** by orchestrating 4 LLMs in a **LangGraph** multi-agent pipeline with ensemble-consensus majority-voting and a zero-redeployment model gateway.
- Lifted classification accuracy **40%** and cut latency **55% (multi-page) / 30% (single-page)** by designing an async parallel inference layer with per-model cost control and confidence-score routing.
- Achieved full production observability with **zero unhandled errors** across 9 failure scenarios by building **Watchtower** (agentic observability UI) and an **MCP** server for governed live tool execution, audited end-to-end via **OpenTelemetry / Datadog**.
- Cut team onboarding time **60%+** by authoring system architecture documentation and running structured **LangGraph / LangChain** KT sessions for 4 engineers.

Platform Engineer

Aug 2023 - Aug 2025

Accenture, Security, Gurugram | DevSecOps & Production Infrastructure

- Achieved **zero customer-facing downtime** during a live pipeline upgrade by orchestrating blue-green deployment across **Route 53, ASG, DynamoDB** and resolving Terraform state corruption via automated Jenkins DR pipelines across 10+ environments.
- Eliminated SIEM telemetry blind spots, measured by **5-min health check coverage** across all enterprise clients, by configuring **mTLS / OTel gRPC** streaming on multi-tenant EC2 and deploying self-healing daemon scripts and **PII log-redaction engines** for global compliance.
- Reduced runtime failures **50%** and closed AppSec vulnerabilities by migrating credentials to **AWS Secrets Manager**, modernizing **12+ Java & Python microservices** (3.8 to 3.12), and building Squid Proxy routing for air-gapped enterprise environments.

PROJECTS

IncidentIQ: AI-SRE Platform [\[GitHub\]](#) | [LangGraph](#) | [FastAPI](#) | [pgvector](#) | [JWT/RBAC](#)

- 5-agent **LangGraph** pipeline: triage, context enrichment, RAG knowledge, RCA, supervision.
- Validated end-to-end: alert ingestion, confidence-scored diagnosis, human-gated rollback, execution with full audit logging.

RAGGym: RAG Learning & Evaluation [\[Live Demo\]](#) | [LangGraph](#) | [Llama 3.2](#) | [Qdrant](#) | [RAGAS](#)

- Swappable RAG architecture: LLM, embeddings, vector store **hot-configurable, no code changes**; dense retrieval + RRF, multi-query, reranking, corrective self-correction loop.
- Supports **Llama 3.2** locally via Ollama or any API model; deployed live with **RAGAS** evaluation harness and AI-graded practice mode.

TECHNICAL SKILLS

GenAI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, MCP, HITL, FSM, Prompt Engineering, NLP, RAG

LLM & Eval: GPT-4o/5, Claude Sonnet, Llama 3.2 (Ollama), Azure AI Foundry, RAGAS, Ensemble Consensus, LLM Benchmarking

Data & Security: PostgreSQL, SQL, pgvector, Qdrant, ChromaDB, CTEs, OAuth 2.0, mTLS, AWS Secrets Manager, PII Redaction

Infrastructure: Python (async), FastAPI, Java, Docker, Terraform, AWS (Lambda, EC2, S3, ASG, Route 53), Azure AKS, GitHub Actions, CI/CD

CERTIFICATIONS

[Claude Certified Architect, Foundations](#)

Anthropic | May 2026

EDUCATION

B.Tech, Electronics and Communication Engineering

Jul 2019 - Aug 2023

Vellore Institute of Technology, Tamil Nadu, India